

ALLWORKSS AUTONOMOUS SYSTEMS INTELLIGENCE

Engineering Research, Deployment & Architecture Report

Enterprise: YARSA ALLWORKSS (OPC) PRIVATE LIMITED	Date: Thursday, August 20, 2026
Brand: AllWorksss	Session Window: 01:30 AM – 03:45 AM IST (~2h 15m)
Lead Architect: Yasar Intakhab Khan	Status: Production Ready / 100% Deployed
Location: Thane, Maharashtra, India	Workspace: F:\AllWorksss_AI_Bridge

1. Executive Summary & Project Purpose

This session accomplished the end-to-end upgrade of the **GRIF-Core Autonomous AI Bridge** (`agent_core.py`). The system fuses local computing power, screen vision, neural voice interaction, and intelligent task routing without relying on external cloud APIs. The resulting framework delivers an air-gapped, zero-latency autonomous engine tailored for enterprise consulting, data operations, workflow automation, and deep logical analysis.

2. Multi-Tier Model Architecture & Routing Matrix

Tier	Model Name	Footprint	Primary Role & Triggers	Latency
Fast Tier	qwen2.5:1.5b	986 MB	File operations, GUI typing triggers, quick chat, app launch	~1.5 s
Reasoning Tier	deepseek-r1:1.5b	1.1 GB	Chain-of-thought math, trading logic, algorithmic flows	3.0 - 6.0 s
Architect Tier	grif-core (7B)	4.7 GB	Full-stack code, FastAPI/Next.js systems, architecture	5.0 - 12.0 s
Embedding Tier	omonic-embed-text	274 MB	Semantic similarity search, offline vector RAG on F:\	< 0.5 s

3. Engineering Implementations & Core Modules

- **Dual-Mode GUI Automation:** Supports live visual interaction (Notepad file display via direct OS process) and instant silent background execution inside the F:\ sandbox.
- **Smart Fuzzy Path Resolver:** Eliminates extension-based false matches using strict 0.75 difflib cutoff logic, distinguishing existing files from new creations.
- **Screen Capture & Vision/OCR Pipeline:** Captures active display frames to temp_screen.png and extracts text/layout via PyTesseract for real-time diagnostics.
- **Natural Neural Voice Synthesis:** Implemented hi-IN-SwaraNeural female voice stream via async Windows Media Player API (`$player = New-Object System.Windows.Media.MediaPlayer`) avoiding Python 3.14 C-compilation blocks.
- **Native App Launcher:** Sub-second subprocess dispatch for Chrome, Notepad, Calculator, Explorer, and Command Prompt.

4. Architecture & Hardened Diagnostic State

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| Host RAM: 15.69 GB Total (6.82 GB Free) | Baseline CPU Load: 7.1% |
| Storage: F:\ Drive Verified (660+ GB Free) | Model Engines: 4 / 4 Active |
| Audio: Async WMP Media Player | Keyboard Interrupt: Clean Ctrl+C Handled |
| Launcher Shortcut: C:\Users\Admin\Desktop\Start_GRIF_Brain.bat |
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5. Strategic & Enterprise Utility for AllWorksss

1. **Air-Gapped Confidentiality:** Zero proprietary data or client records leave the workstation.
2. **Rapid Multi-Tasking:** Routine file diagnostics and automation execute in < 1.5s via sub-model routing.
3. **Multi-Modal Intelligence:** Native Indian neural voice and screen OCR create a hands-free workflow.
4. **Local RAG Foundation:** Semantic vector indexing prepares the system for deep internal enterprise search.